



Introduction to Python

Agenda

- Introduction
- Why Python?
- Simple Expressions
- Variables
- Basics of Data Types
- Operators
- Conditional Statements and Loops

- Python is a high-level programming language and was created by Guido van Rossum
 - Readability
 - Ease of Learning
 - Large Standard Library
 - Interpreted and Interactive
- Python is also known as a general-purpose programming language, as it is used in the various domains
 - Web Development
 - Software Development
 - Game Development
 - AI & ML
 - Data Analytics

Why Python?

- Python works on different platforms (Windows, Mac, Linux)
- Syntax used is simple to understand and code
- Supported by a large community
- Contains many libraries that can be used for data science tasks, on top of some standard libraries written in Python
- It has the feature of dynamic typing, where the data type of the variable is determined only during runtime

Libraries in Python

- The Python standard library consists of more than 200 core modules
- Standard Libraries for Data Science
 - NumPy (Numerical Computations)
 - Pandas (Data frames)
 - Matplotlib (Visualization)
 - Statsmodels (Statistics)
 - Scikit-learn (Machine Learning)

Coding environment -IDE

- Integrated Development Environment (IDE) is a software product solely developed to support software development in various or specific programming language(s)
- An IDE usually comprises of
 - Source code editor
 - Compiler
 - Debugger
- Additional features include syntax and error highlighting, code completion


```
operation == "MIRROR_X":  
    mirror_mod.use_x = True  
    mirror_mod.use_y = False  
    mirror_mod.use_z = False  
operation == "MIRROR_Y":  
    mirror_mod.use_x = False  
    mirror_mod.use_y = True  
    mirror_mod.use_z = False  
operation == "MIRROR_Z":  
    mirror_mod.use_x = False  
    mirror_mod.use_y = False  
    mirror_mod.use_z = True
```

```
#selection at the end -add  
mirror_ob.select= 1  
modifier_ob.select=1  
context.scene.objects.active  
= ("Selected" + str(modifier_ob.name))  
mirror_ob.select = 0  
= bpy.context.selected_objects  
data.objects[one.name].select  
print("please select exactly one mirror")
```

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```
def mirror(modifier):  
    #add mirror to the selected  
    #object -mirror_x  
    mirror_x
```

THANK YOU